Ravindra Chowdary Jonnagadla

AI Systems Engineer • Machine Learning Researcher

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Professional Experience

Waani – Conversational Voice AI Platform — Chennai, India

Founder & Lead Engineer — July 2026 – Present

- Architected an ultra-low-latency duplex voice pipeline orchestrating telephony protocols, streaming STT (Sarvam AI), and LLM token inference, achieving **<800ms p95 voice-to-voice turn-around time** across 5,000+ concurrent voice sessions.
- Engineered conversational state machines and dynamic intent branching for automated lead qualification, instrumenting Prometheus metrics and Grafana dashboards to track streaming audio packet jitter and queue depth.
- Built an asynchronous dual-channel audio ring buffer with edge Voice Activity Detection (VAD) over WebSockets, implementing atomic cancellation tokens to flush in-flight TTS buffers on instant user barge-in.

LynkZen – Instagram DM Automation SaaS — Chennai, India

Founder & Lead Engineer — April 2026 – June 2026

- Built and launched an event-driven Instagram automation engine using Next.js, TypeScript, and Tailwind CSS, decoupling webhook ingest via BullMQ asynchronous worker queues to absorb bursts of 10k+ concurrent webhook events.
- Developed an interactive node-based visual workflow builder allowing creators to configure multi-step conditional messaging funnels and automated response paths without writing code.
- Scaled platform to **3,000 active beta signups for closed testing**, implementing token-bucket rate limiting via atomic Redis Lua scripts to eliminate race conditions and guarantee zero 429 quota violations against Meta Graph API limits.

Technical Projects & Research

AMERS: Multimodal Emotion Recognition System • *PyTorch, Signal Processing* — **Hugging Face** • **IEEE ICETEA 2026**

- Developed a multimodal emotion recognition architecture synchronizing multi-channel EEG (128Hz) and speech spectrograms via Cross-Modal Mutual Attention (CMMA) to capture inter-modal temporal-spatial dynamics.
- Integrated an Emotion-Aware Gating (EAG) WGAN-GP generator with a PPO reinforcement learning controller, dynamically scheduling synthetic-to-real training batches against discriminator reward stability.
- Evaluated on DEAP/SEED benchmarks via LOSO cross-validation across 4 emotion classes, achieving **82.55% accuracy** (+7.4% over baselines), **0.83 macro F1-score**, and **0.77 Cohen's kappa**, open-sourcing checkpoints on **Hugging Face**.

SmartCap: Automated Subtitle & Transcoding Pipeline • *Python, Whisper, FFmpeg, ONNX* — **IEEE ICICV 2025**

- Built an automated transcription pipeline combining Whisper with timestamp alignment for frame-accurate subtitles (.srt/.vtt), quantizing to INT8 via ONNX Runtime to achieve a **0.35x Real-Time Factor (RTF)**, published as Lead Author (Doc ID: 11085491).
- Architected an asynchronous worker pipeline orchestrating FFmpeg stream filters for dynamic audio normalization and hardcoded subtitle burning, isolating CPU-bound transcoding to eliminate event-loop starvation.

Smart QR-Based University Parking System • *Flask, Python, PostgreSQL, Redis* — **IEEE CONIT 2025**

- Designed a campus parking management application with HMAC-SHA256 signed dynamic QR codes, timestamped nonces cached in Redis to prevent replay attacks, and Twilio OTP verification, published as Co-Author (Doc ID: 11166870).
- Engineered slot reservation concurrency using optimistic row-level locking with versioning in PostgreSQL, maintaining **zero double-booking anomalies and a 99.8% transaction success rate** under simulated 1,000-user rush-hour contention.

Technical Skills

Languages: Python, C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, Bash, Shell Scripting

AI/ML & Research: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Multimodal Learning, WGAN-GP, Reinforcement Learning, PPO, NLP, EEG Signal Processing, Vector Search, ChromaDB

Generative AI & Speech: Whisper, Faster-Whisper, ONNX Runtime, Model Quantization, Sentence Transformers, RAG, FFmpeg, Audio/Speech Processing (STT/TTS)

Backend & Systems: FastAPI, Flask, Django, Node.js, Next.js, React, RESTful APIs, WebSockets, Redis, BullMQ, Celery, Microservices

Cloud, DevOps & Observability: AWS (EC2, S3, Lambda, IAM), Cloudflare R2, Docker, Kubernetes, Terraform, GitHub Actions, Linux, Git, Prometheus, Grafana

Certifications: Oracle Certified Professional: MySQL 8.0 Database Developer (1Z0-909) • AWS Certified Cloud Practitioner

Education

SRM Institute of Science and Technology — Chennai, India

B.Tech in Computer Science and Engineering – Cloud Computing (CGPA: 8.01/10) — May 2026